
EXERCISES

Homework 1

AI502: Ethics and Privacy

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2026



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1 Problem 1

What are the domain and range of the following functions?

1. $g(x) = \frac{1}{x}$
2. $f(x) = \sqrt{x-3}$

1.1 Solution

1. since $g(x) = \text{DNE}$ when $x = 0$

Therefore the domain is $x \in \mathbb{R} \setminus \{0\}$

2. I assume that $f(x)$ is only defined in non-complex numbers (not i)

The domain is therefore: $x \in [3, \infty)$

2 Problem 2

Let

$$f(x) = \begin{cases} x-5 & x \geq 1, \\ -3x & x < 1 \end{cases}$$

Evaluate $\lim_{x \rightarrow 1} f(x)$

2.1 Solution

To evaluate the limit check right hand limit $x \rightarrow 1^+$ and left hand limit $x \rightarrow 1^-$

$$\lim_{x \rightarrow 1^+} f(x) \rightarrow 1 - 5 = -4$$

$$\lim_{x \rightarrow 1^-} f(x) \rightarrow -3 \cdot 1 = -3$$

And since $-4 \neq -3$ the limit $\lim_{x \rightarrow 1} f(x)$ **does not exist**

3 Problem 3

Compute:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

3.1 Solution

On the surface, this looks complicated but since

$$\frac{x^2 - 4}{x - 2} = \frac{x^2 + 2x - 2x - 4}{x - 2} = \frac{(x-2)(x+2)}{x-2} = x+2$$

So actually

$$\lim_{x \rightarrow 2} x + 2 = 4$$

4 Problem 4

Evaluate

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$$

4.1 Solution

Since $\frac{\sin(5 \cdot 0)}{0}$ is undefined. Use L'Hôpital's rule that is

Theorem: L'Hôpital's Rule

Let f and g be functions that are differentiable on an open interval I containing a , except possibly at a itself

Assume that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm\infty$$

Then also assume that $g'(x) \neq 0, \forall x \neq a$

The the limit is

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

by theorem

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{x} = \lim_{x \rightarrow 0} \frac{\cos(5x)}{1} = \frac{5 \cdot \cos(0)}{1} = 5$$

5 Problem 5

Determine whether

$$f(x) = \begin{cases} \frac{\sin(x)}{x} & x \neq 0, \\ 1 & x = 0 \end{cases}$$

is continuous at $x = 0$

5.1 Solution

Yes it is, since $f(0) = 1$

6 Problem 6

Find all points where $g(x) = \frac{x^2-1}{x-1}$ is continuous.

6.1 Solution

Since $g(0)$ is undefined

This is continuous on all points except 1, so $(-\infty, 1) \cup (1, \infty)$

7 Problem 7

Differentiate

1. $f(x) = x^3 \sin(x)$
2. $g(x) = \ln(\sqrt{1+x^2})$

7.1 Solution

$$\frac{d}{dx} ax^n + b$$

1. Use product rule

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

so..

$$\frac{d}{dx} f(x) = 3x^2 \cdot \sin(x) + x^3 \cdot \cos(x)$$

2. Use chain rule

$$\frac{d}{dx}(f(g(h(x)))) = f'(g(x)) \cdot (g'(h(x)) \cdot h'(x))$$

so..

$$\frac{d}{dx} g(x) = \frac{1}{\sqrt{1+x^2}} \cdot \left(\frac{1}{2}(1+x^2) \cdot 2x \right)$$

8 Problem 8

Find the equation of the tangent line to

$$y = e^{2x}$$

at $x = 0$

8.1 Solution

Use chainrule again

$$\frac{d}{dx}e^{2x} = e^{2x} \cdot 2$$

at $x = 0$

$$a = 2$$

This is the slope of the tangent

Now find the tangent intersect at $x = 0$

$$y(0) = e^{2 \cdot 0} = 1$$

So point $(0, 1)$ is the intersect

Now for the tangentline $y - y_0 = a(x - x_0)$

This equals to

$$y - 1 = 2(x - 0) \implies y = 2x + 1$$

9 Problem 9

Consider $f(x) = (x^2 - 4)^7$. Find and classify all local extrema

9.1 Solution

Use chainrule Find $f'(x) = 0$

$$0 = 7(x^2 - 4)^6 \cdot 2x = 14x(x^2 - 4)^6$$

Use 0 rule. either $14x$ or $(x^2 - 4)^6$ equals 0 for the result to equal 0

So..

$$14x(x^2 - 4)^6 \Rightarrow x = 0 \quad \text{or} \quad x^2 - 4 \Rightarrow x = \pm 2$$

So critical points are $= -2, 0, 2$.

10 Problem 10

Find the global max and min of $f(x) = x^3 - 7x + 6$ on $-4 \leq x \leq 2$.

10.1 Solution

For this find $f'(x)$ and choose max and min.

$$f'(x) = 3x^2 - 7$$

Now solve for $f'(x) = 0$

$$x = \pm\sqrt{\frac{7}{3}}$$

So see which of these are the max and min

$$f''(x) = 6x \Rightarrow 6 \cdot \sqrt{\frac{7}{3}} > 0 \quad \text{and} \quad 6 \cdot \left(-\sqrt{\frac{7}{3}}\right) < 0$$